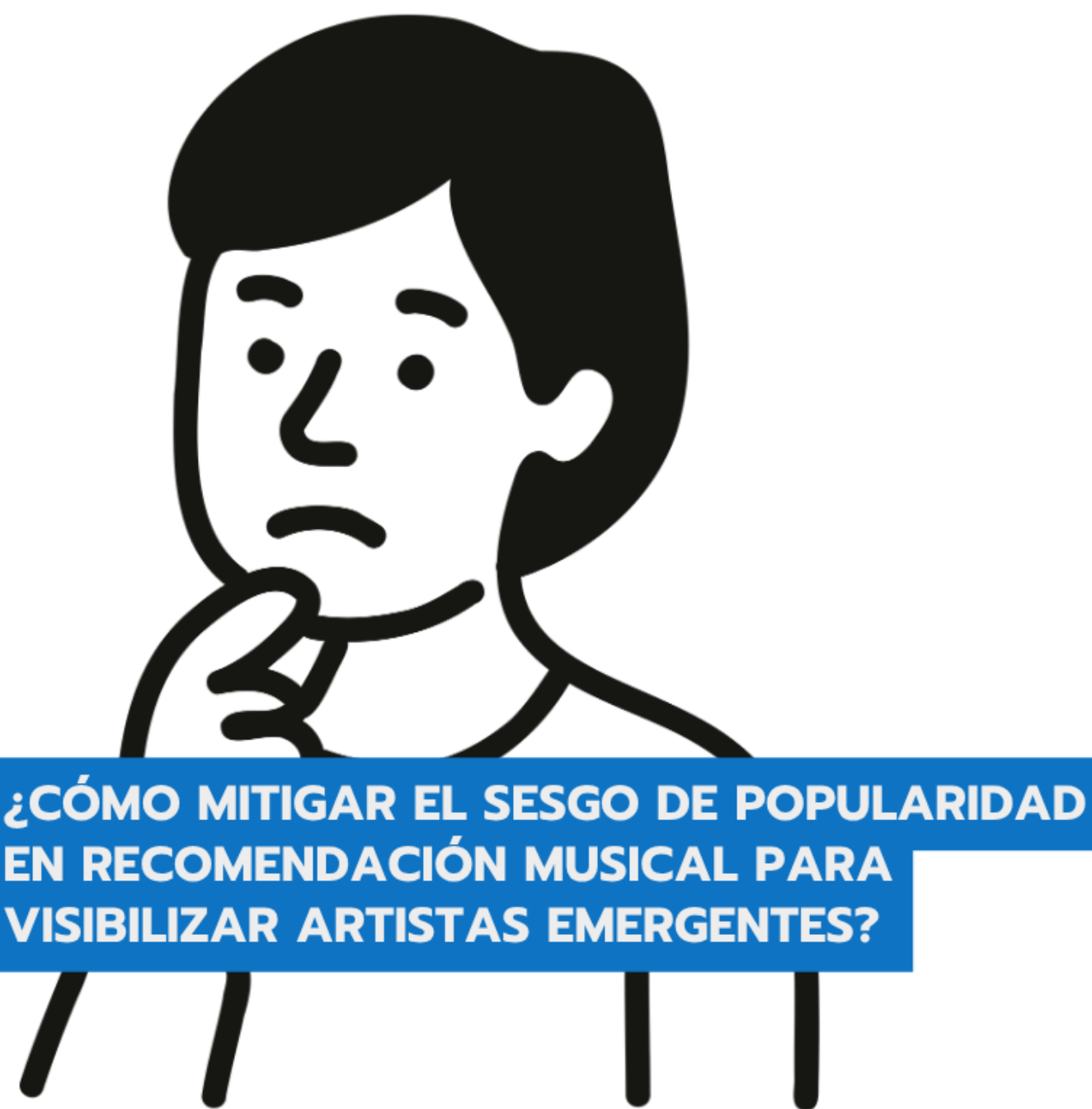


1. PROBLEMA: EL SESGO DE POPULARIDAD

"Music recommendation systems face a persistent challenge: emerging artists struggle to gain visibility because most algorithms prioritize popularity signals over content affinity, creating feedback loops that concentrate attention on established hits."

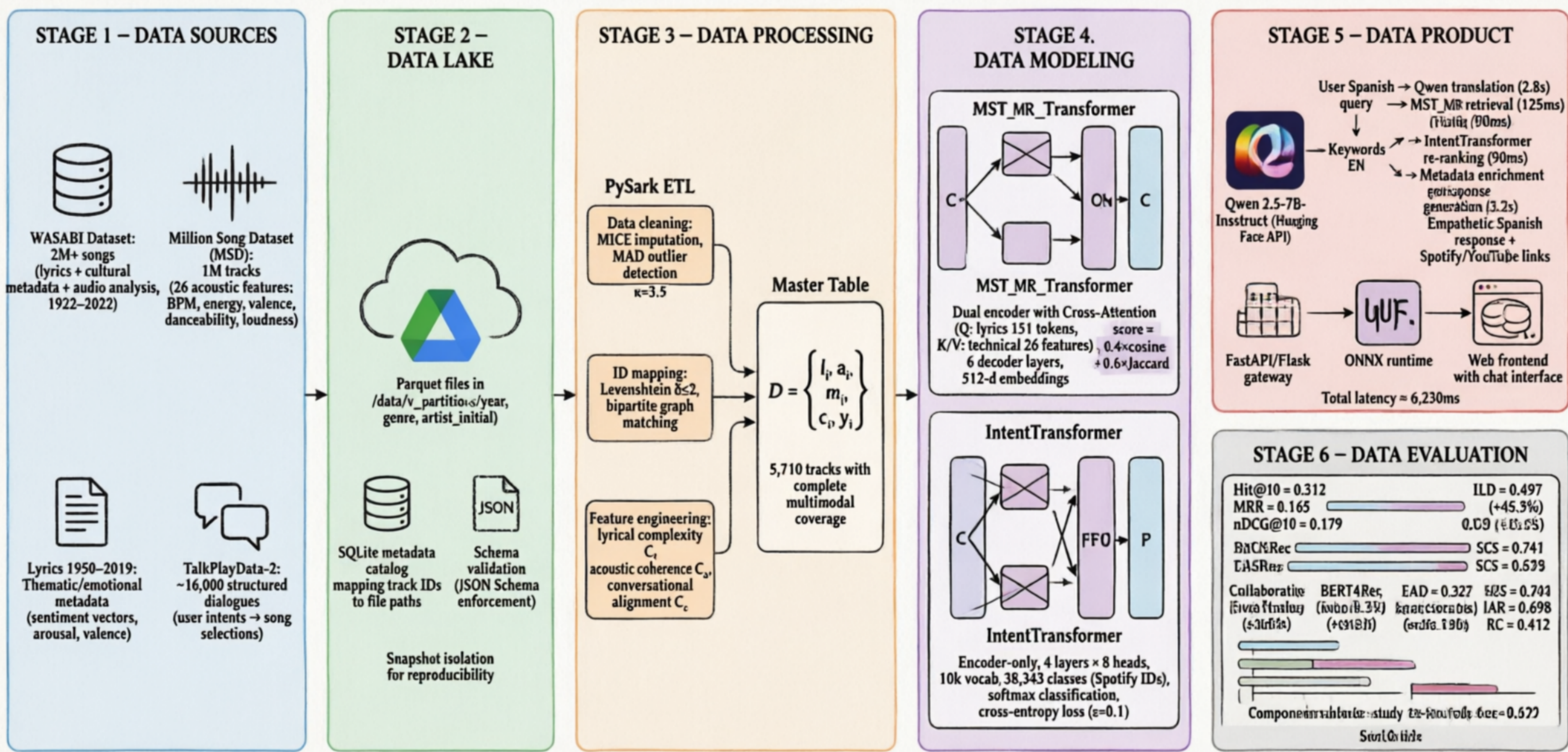


2. OBJETIVO GENERAL

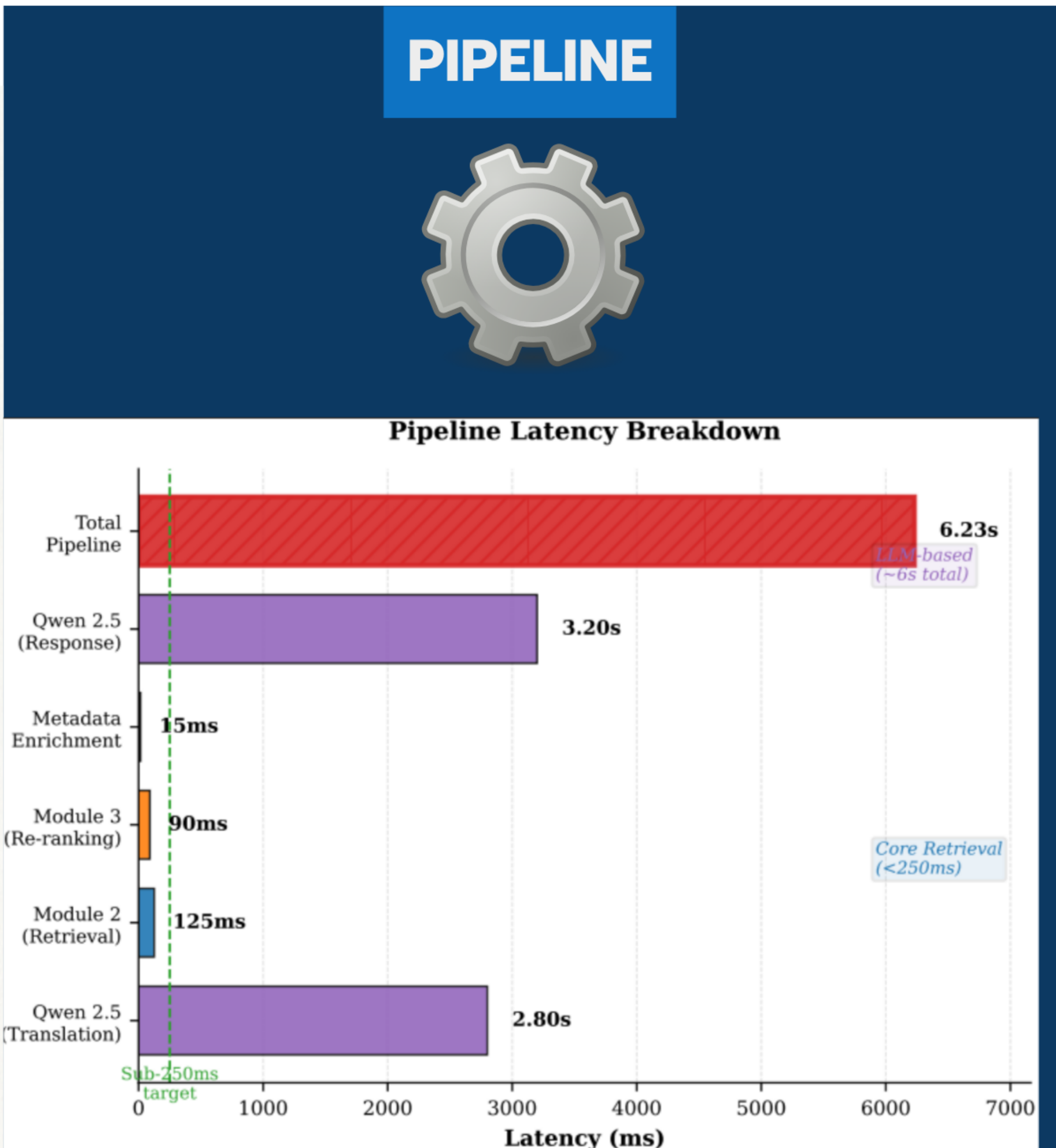
"Desarrollar un agente recomendador de música híbrido que integre técnicas tradicionales de recomendación, arquitectura de transformers y modelos de lenguaje generativo (LLM) para una interfaz conversacional; adaptándose dinámicamente a las preferencias y retroalimentación de los usuarios."

ARQUITECTURA

EchoPulse AI Multimodal Music Recommender System Architecture

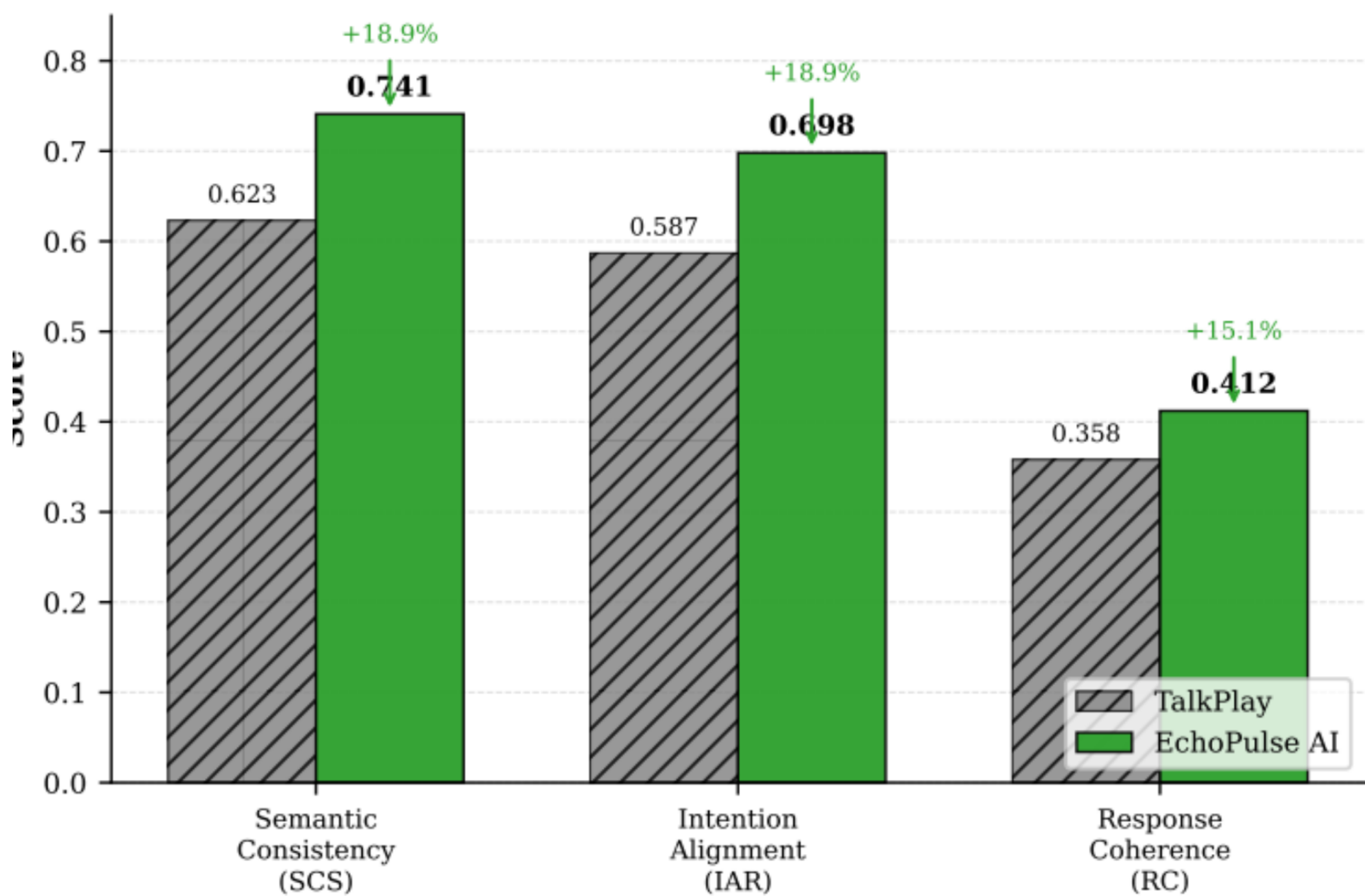


PIPELINE

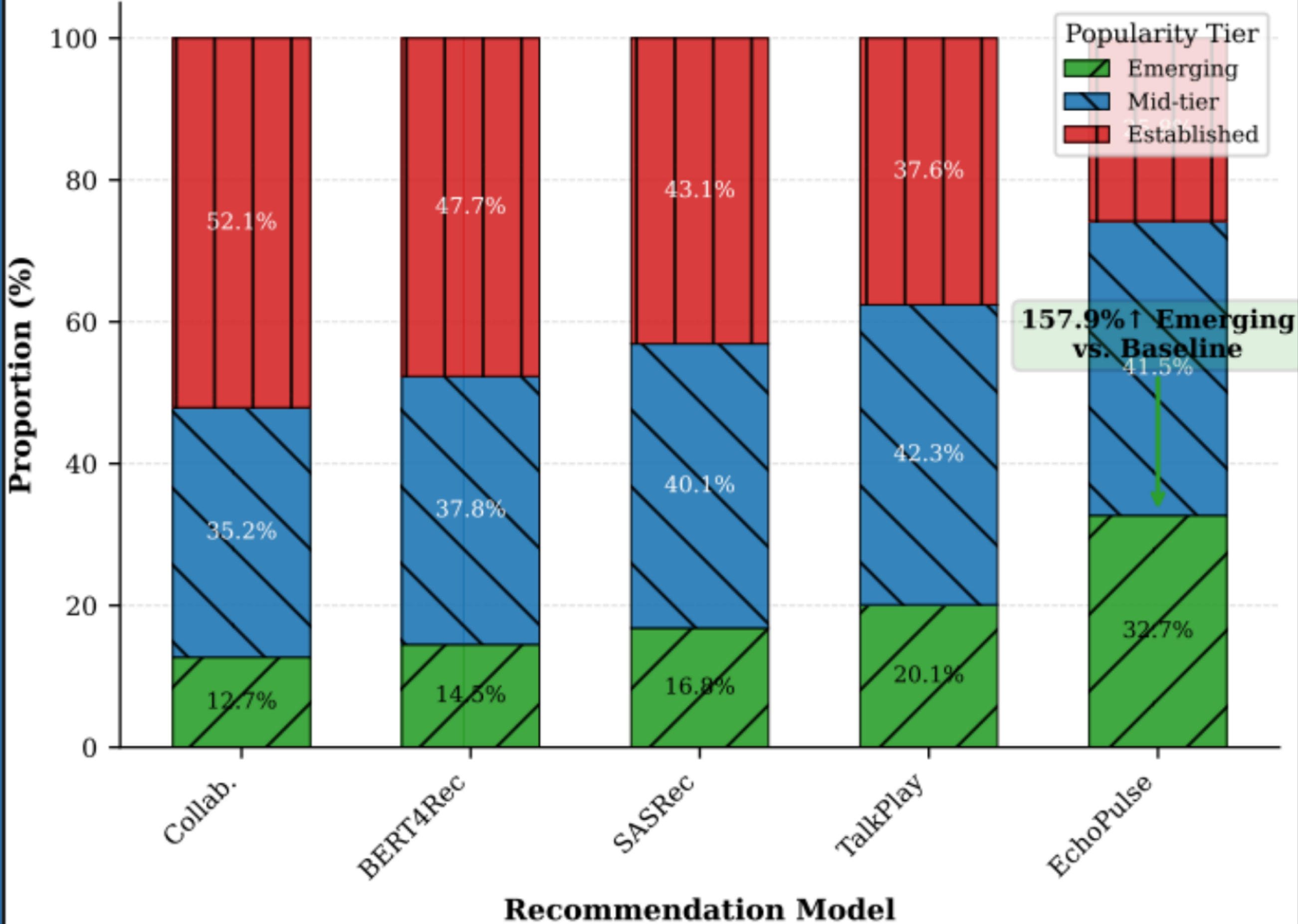


RESULTADOS

Generative Transformer Evaluation



Distribution of Recommended Artists by Popularity



1. VASWANI, A., SHAZEER, N., PARMAR, N., USZKOREIT, J., JONES, L., GOMEZ, A.N., KAISER, L., POLOSUKHIN, I.: ATTENTION IS ALL YOU NEED. IN: ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 30 (NEURIPS), PP. 5998–6008 (2017)

3. KANG, W.C., MCAULEY, J.: SELF-ATTENTIVE SEQUENTIAL RECOMMENDATION. IN: IEEE INTERNATIONAL CONFERENCE ON DATA MINING (ICDM), PP. 197–206 (2018)

Model	Hit@10	MRR	nDCG@10	ILD	EAD	SCS	IAR
Collaborative Filtering	0.142	0.089	0.156	0.342	0.127	0.412	0.385
BERT4Rec [4]	0.158	0.095	0.172	0.389	0.145	0.438	0.401
SASRec [3]	0.164	0.102	0.178	0.412	0.168	0.451	0.423
TalkPlay [8]	0.189	0.118	0.165	0.445	0.201	0.623	0.587
EchoPulse AI	0.312	0.165	0.179	0.497	0.327	0.741	0.698

2. DEVLIN, J., CHANG, M.W., LEE, K., TOUTANOVA, K.: BERT: PRE-TRAINING OF DEEP BIDIRECTIONAL TRANSFORMERS FOR LANGUAGE UNDERSTANDING. IN: PROCEEDINGS OF NAACL-HLT, PP. 4171–4186 (2019)

BIBLIOGRAFÍA